UNIVERSITY OF MINES AND TECHNOLOGY, TARKWA

SECOND SEMESTER EXAMINATION, AUGUST 2024

COURSE NO: EL/CE 464
COURSE NAME: INTRODUCTION TO ROBOTICS
CLASS: **EL/CE IV** **TIME: 2$^1/_2$HRS**

Name**:** ____Index Number: ________________

SECTION A

Circle the Correct Answer In This Section On The Question Paper

1. What distinguishes Continuous Path Robots from Point-to-Point Robots?
a) Continuous Path Robots operate with discrete movement steps, while Point-to-Point Robots move along a path without stopping.
b) Continuous Path Robots move along a continuous path, allowing for smooth, uninterrupted motion, while Point-to-Point Robots move between specific points.
c) Continuous Path Robots have a higher payload capacity than Point-to-Point Robots.
d) Continuous Path Robots are used for assembly tasks, while Point-to-Point Robots are used for pick-and-place tasks.

2. In robotic arms, which joint type is typically used to create a twisting motion?
a) Revolute joint
b) Spherical joint
c) Planar joint
d) cylindrical joint

3. A classic example of servo -controlled robot(s) is/are…………………………………………..
I. Seiko PN-100
II. Ultimate 2000
a) I=True , II= True
b) I=True, II=False
c) I=False , II=True
d) I=False , II=False

4. All the following are mobile robots except ……………………………………………….
a) Tracked Robots
b) Wheeled Robots
c) Multi-legged Robots
d) Manipulators

5. Which statement best explains dextrous workspace?
a) It is the volume of space that the end-effector can reach with a minimum of one orientation
b) It is the volume of space, which the robot's end-effector can reach with various orientations
c) Dextrous workspace is a comprise the reachable workspace
d) It is the volume of space that the end-effector can reach with a maximum of one orientation

6. In a prismatic joint, which parameter is variable?
a) θ_i
b) d_i
c) θ_{i-1}
d) d_{i-1}

7. In which industry are robots most used for tasks like welding, painting, and assembly?
a) Agriculture
b) Healthcare

c) Education
d) Manufacturing

8. Which type of manipulator is considered redundant?
a) A Spatial Manipulator with 6 DOF
b) A Planar Manipulator with 3 DOF
c) A Spatial Manipulator with more than 6 DOF
d) A Planar Manipulator with less than 3 DOF

9. What is the primary focus of path planning in robotics?
a) Determining the speed and timing of the robot's movements
b) Finding a collision-free path in the configuration space (C-space)
c) Analyzing the robot's kinematic constraints
d) Computing the robot's energy consumption

10. Under what conditions does a company decide to purchase the robot?
a) Pay-back period is greater than the techno-economic life, and rate of return on investment is less than the rate of bank interest
b) Pay-back period is greater than the techno-economic life, and rate of return on investment is greater than the rate of bank interest
c) Pay-back period is less than the techno-economic life, and rate of return on investment is greater than the rate of bank interest
d) Pay-back period is less than the techno-economic life, and rate of return on investment is less than the rate of bank interest

11. Which type of end effector is ideal for picking up metal objects?
a) Soft gripper
b) Magnetic gripper
c) Vacuum suction cup
d) Cutting tool

12. If a robot's Basic Resolution Unit (BRU) is defined as 0.01 inch, what does this signify?
a) The robot can lift objects up to 0.01 inch in size
b) The robot's accuracy is limited to 0.01 inch
c) The robot can make position adjustments as small as 0.01 inch
d) The robot can move at a speed of 0.01 inch per second

13. The main difference between active gripper and passive gripper is ……………………………
a) Active gripper relies on mechanical design and physical interactions and passive gripper relies on electricity or hydraulics
b) Active gripper has sensors and passive gripper is without sensors
c) Active gripper has unprecise control over the gripping force and passive gripper has precise control over the gripping force.
d) Active gripper operates manually while passive gripper operates electronically

14. What are the two main types of robotic kinematics?
a) Linear and rotational kinematics
b) Forward and inverse kinematics
c) Static and dynamic kinematics
d) Discrete and continuous kinematics

15. In a robotic arm with 6 degrees of freedom, how many independent joint variables are there?
a) 3
b) 4
c) 5
d) 6

16. What does the Denavit-Hartenberg (D-H) convention help with in robot kinematics?
a) It is a method for robot vision processing.
b) It provides a standardized way to represent joint parameters.
c) It measures the temperature of robotic components.
d) It controls the speed of the robot's motors.

17. What does a kinematic chain represent in robotics?
a) A series of linked joints and segments that define the structure of a robot
b) The chain reaction that powers the robot
c) The sequence of commands executed by the robot
d) The communication protocol between robot sensors

18. Which of the following is typically used to describe the orientation of a robot's end effector?
a) Cartesian coordinates
b) Euler angles
c) Voltage levels
d) Resistance values

19. In a robotic system, what is the purpose of using homogeneous transformation matrices?
a) To calculate joint torques
b) To solve inverse kinematics problems
c) To control the speed of the robot
d) To describe the position and orientation of a frame relative to another frame

20. What is the primary goal of Reactive Control Strategies in motion planning?
a) To create a static roadmap of the environment
b) To adaptively respond to changes in the environment in real-time
c) To pre-compute paths before the robot starts moving
d) To divide the environment into discrete cells

21. How do you transform a point P from frame A to frame B using the homogeneous transformation matrix T_{AB}
a) Multiply P by the rotation matrix only
b) Add the translation vector to P
c) Multiply P by the homogeneous transformation matrix $\boldsymbol{T_{AB}}$
d) Subtract the translation vector from P

22. If a robot's orientation needs to be changed to face a different direction while keeping its body level, which angle is primarily adjusted?
a) roll
b) pitch
c) yaw
d) elevation

23. Trajectory planning includes which of the following considerations?
a) Generating a static path in the environment
b) Mapping the robot's environment
c) Avoiding static obstacles in the configuration space
d) Determining the time-parameterized path for the robot

24. Which components affect the torque in a robotics joint?
I. Inertia

II. Centrifugal/Coriolis
III. Friction
IV. Gravity

a) I and II
b) II and III
c) I, II and III
d) I, II and IV

25. Match correctly the characteristics of the sensors

A. Range I. Deviation from the exact quantity
B. Sensitivity II. Constant sensitivity
C. Linearity III. Change in output/change in input
D. Accuracy IV. Difference between maximum and maximum values of measured input

a) A=II, B=III, C=IV and D=I
b) A=III, B=IV, C=II and D=I
c) A=I, B=II, C=IV and D=III
d) A=IV, B=III, C=II and D=I

26. What is the primary advantage of using Linear Variable Differential Transformer over a potentiometer for displacement measurement?
a) Higher sensitivity
b) Greater accuracy and linearity
c) Lower cost
d) Smaller size

27. In the Traditional Scheme of motion planning, which method is used to break up the environment into simpler regions?
a) Potential Field
b) Cell Decomposition
c) Probabilistic Approach
d) Incremental Planning

28. Which sensor type is based on the principle of Lorentz force?
a) Capacitive Sensor
b) Inductive Sensor
c) Half effect Sensor
d) Motion Sensor

29. Which of the following best describes a typical application of frame grabbing technology?
a) Monitoring the speed of rotating machinery
b) Measuring ambient temperature in a room
c) Detecting sound frequency in an audio signal
d) Capturing high-resolution images from a live video feed for inspection or analysis

30. What technique is used to separate objects from the background in an image?
a) Image compression
b) Image thresholding
c) Image filtering
d) Image scaling

31. In what kind of application would a robot vision system that combines both RGB and depth cameras be most beneficial?
a) In an environment where precise object shape and distance information are required for interaction, such as in autonomous navigation or robotic surgery
b) In an environment where only temperature readings are needed
c) In an environment where only audio feedback is used
d) In an environment where only basic image capturing is required

32. Which Analytical Approach in Traditional Scheme involves guiding the robot using forces derived from the environment?

a) Path Velocity
b) Potential Field
c) Relative Velocity
d) Reactive Control Strategies

33. What is a key feature of fuzzy logic in non-traditional motion planning schemes?

a) It provides precise numerical values for decisions.
b) It uses binary logic to make decisions.
c) It allows for reasoning with imprecise and uncertain information.
d) It only works in static environments.

34. In the robot walking cycle, what is the phase where both feet are in contact with the ground called?

a) Single Support Phase
b) Double Support Phase
c) Swing Phase
d) Stance Phase

35. The term that describes the point where the resultant of the robot's dynamic forces and moments cancels out, preventing tipping over is ……………………………………………………………….

a) Zero Magnitude Point
b) Zero Motion Point
c) Zero Moment Point
d) Zero Mechanical Power

36. Which of the following is a key consideration when designing hip joint trajectories for stair climbing robots?

a) Battery life optimization
b) Maintaining a constant speed
c) Minimizing sensor data usage
d) Ensuring that the trajectory accommodates the step height and depth

Use the figure to answer question 37 and 38

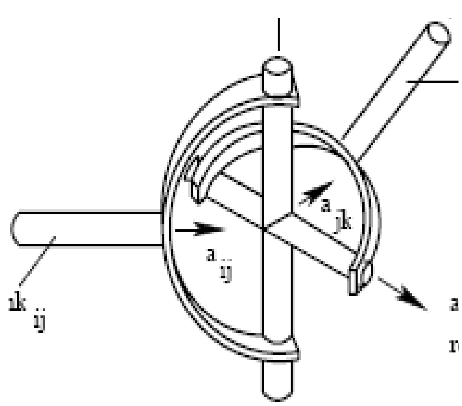

Figure 1

37. Identify the type of joint in figure 1

a) Revolute Joint
b) Prismatic Joint
c) Cynderical Joint
d) Hooke Joint

38. What is the primary purpose of a Hooke's joint?
a) To transmit torque between two shafts with a constant speed ratio
b) To transmit torque between two shafts with varying angular speeds
c) To connect two shafts that must remain perfectly aligned
d) To reduce friction between two rotating shafts

39. The Grubler's criterion applies primarily to which type of mechanisms?
a) Simple planar mechanisms
b) Complex spatial mechanisms
c) Single-degree-of-freedom mechanisms
d) Gear trains

40. What unique ability does a robotic system with the "TRP" or "TRS" configuration have?
a) High-speed processing
b) Handling delicate objects with precision
c) Picking up objects lying on the floor
d) Operating in extreme temperatures

SECTION B

Attempt Question **One** and any other Questions from This Section into Your Answer Booklet

Question One:

A manufacturing company uses robotic arms to automate the assembly line process. One of the tasks involves moving a part from one location to another along a predefined path. To ensure smooth and efficient movement, the robotic arm must follow a cubic trajectory. As a robot expert, you are tasked with planning this trajectory using a cubic polynomial.

The robotic arm needs to move a part from an initial position $p(0)$ to a final position $p(t)$within a specified time duration T. The movement should start and end with zero velocity to avoid sudden jerks that could destabilize the part or damage the machinery.

Given a general cubic polynomial as:

$$p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$$

The velocity $\dot{p}(t)$is the first derivative of $p(t)$ with respect to t

$$p(t) = a_1 + 2a_2 t + 3a_3 t^2$$

Boundary Conditions given as:

Initial position: $p(0) = 0$
Final position: $p(T) = 1$
Initial velocity: $\dot{p}(0) = 0$
Final velocity: $\dot{p}(T) = 0$

a) Solve the cubic polynomial that describes the trajectory $p(t)$ of the robotic arm [5 marks]
b) Sketch the graphical representation of the trajectory in a) [1 mark]
c) Briefly explain the four types of polynomial trajectories. [2 marks]
d) Discuss four analytical approaches robot can use to deal with moving obstacles in its environment. [2 marks]

Question Two

a) With the aid of diagram, briefly explain the robot manipulator system. [5 marks]
b) Define a robot end-effector and describe its primary functions in a robotic system. [2 marks]
c) What are the key considerations when selecting an end-effector for a specific robotic application? [2 marks]
d) Briefly explain frame grabbing as related to robot vision [1 mark]

Question Three

a) Give the definition of intelligent robot according to your understanding? [2 marks]
b) State the main difference between free space motion time and complaint motion time in robotic systems. [2 marks]
c) Discuss the importance of integration of manipulation and navigation in robots designed for a delivery system. [2 marks]
d) With the aid of diagram, explain four graph-based approaches used in robot motion planning. [4 marks]

Questions Four

Consider a 2-link robotic arm with three reference frames: Frame 0 (base frame), Frame 1 (attached to the first link), and Frame 2 (attached to the second link). The robotic arm operates in a 2D plane. The following parameters are provided:

Link 1:

- Length L1=4L_1 = 4L1=4 units
- Rotation $\theta_1 = 45^0$=45∘ about the Z-axis
- Translation along the X-axis from Frame 0 to Frame 1

Link 2:

- Length L2=3L_2 = 3L2=3 units
- Rotation $\theta_2 = 43$about the Z-axis
- Translation along the X-axis from Frame 1 to Frame 2

a) Calculate the overall transformation matrix of the robotic arm [3 marks]

b) Describe the position and orientation based on the overall transformation matrix. [3 marks]

c) State the main difference between forward kinematics and inverse kinematics [2 marks]

d) Briefly explain four application of intelligent robots. [2 marks]

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